

1) Tier Classification

**Input
from EMD**

*survived &
candidate_
equivalent
mutants*

Tier 1

L2 divergence seen

Tier 2

Survived after EMD

Tier 3

candidate_eq subset

Execution order

T1 → **T2** → **T3**

2) Deterministic Enhanced Testing

config_stress

batch-size variation { 1, 2, 4, 8, 16, 32, 64 } × 3 seeds

tier1_replay

*Replay recorded L2 divergence
under allclose (Tier 1 only)*

dtype_stress

*FP16 / BF16 casts × 3 seeds · targets
acc_downgrade, cast_remove*

training_stress

*.train() mode × 21 policies × 3 seeds ·
applicable to 5 operators (BN/LN paths)*

value_stress

*15–21 stress policies × 3–5 seeds
(excludes L2-tested)*

repeated_run

*10 reruns × 3 seeds, atol=rtol=1e-6
· targets sync_remove*

3) Final Labeling

Per-mutant

-  **killed**
-  **confirmed_
equivalent**
Tier 3 only
-  **survived**

*first-kill dimension
cross-dim confirmation flag*